Journal of Retailing and Consumer Services 85 (2025) 104307

Contents lists available at ScienceDirect

Journal of Retailing and Consumer Services

journal homepage: www.elsevier.com/locate/jretconser

Consumer behavior toward cultured meat in the foodservice industry: Insights from IPA and fsQCA analysis on shifting trends

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ARTICLE INFO

Keywords:
Cultured meat attributes
Importance-performance analysis (IPA)
Fuzzy-set qualitative comparative analysis (fsQCA)
Future food
Food service industry
Consumer approach behavior

ABSTRACT

Cultured meat is gaining prominence as a food of the future, attracting the attention of researchers and consumers. This study explores the key factors influencing consumer visit intentions for cultured meat restaurants through safety, quality, and ethical value propositions. By conducting interviews with 32 participants and an online survey with 571 respondents, this study employs importance-performance analysis (IPA) and fuzzy-set qualitative comparative analysis (fsQCA) to analyze complex relationships among these factors. The findings identify eight critical attributes of cultured meat that influence consumer decisions, providing strategic insights for restaurant marketers in China. Notably, the study reveals that while novelty and ethical concerns play a role, key determinants such as price, quality, nutritional benefits, and aversion significantly impact visit intentions. The fsQCA results emphasize the importance of combined attributes rather than single factors, highlighting the interplay between consumer perceptions and restaurant success. These results advance the understanding of cultured meat adoption and contribute to both theoretical and practical discussions on consumer behavior in the foodservice industry. Ultimately, this study offers implications for restaurant industry managers and policymakers aiming to effectively develop the cultured meat market.

1. Introduction

Meat plays a crucial role in human nutrition by providing a variety of essential nutrients. Meat consumption is a global phenomenon, which is influenced by cultural, economic, and social factors. According to the Food and Agriculture Organization's updated projections, global meat demand is expected to rise steadily until 2050, posing a significant challenge to livestock systems (Rao et al., 2023). Between 1998 and 2018, global meat consumption, driven primarily by population growth, increased by 58 %, rising from approximately 200 million tons to 360 million tons. Income growth had the greatest impact on per capita meat consumption (Whitnall and Pitts, 2019). Meat consumption in China increased by 72 % between 1998 and 2018, accounting for 34 % of the global meat consumption growth, which was from around 50 kg–79 kg per person (Whitnall and Pitts, 2019). A similar trend occurred in Indonesia, which increased from about 19 kg to 39 kg/per person. Indonesia's strong economic and population growth have contributed to this rise. Furthermore, the growth in meat consumption in Australia increased from around 99 kg–109 kg per capita, and it increased from around 110 kg–119 kg per capita in the United States, which were modest (Australian Government 2020). In addition, many countries in Africa are currently experiencing population growth (Galal, 2024; INTELLINEWS, 2024). Consumer demand for meat will enormously increase as incomes rise in other fast-growing countries, which is similar to how meat consumption has soared in China and Indonesia over the past 20 years.

The overwhelming consumption of meat poses a severe environmental threat (Yu et al., 2023), jeopardizing the Earth's capacity to sustain an anticipated population of 9 billion by 2050, which includes over 90 % of the global populace indulging in meat (Halweil and Nierenberg, 2012). Transitioning the world's population to veganism has proven to be unfeasible due to the widespread love for meat and the inherent difficulty in altering dietary preferences (Rosenfeld, 2018). Advancements in agribusiness have led to the emergence of cultured meat production as a promising solution to address the negative consequences that are linked with conventional meat production in response to these challenges (Rasmussen et al., 2024). Cultured meat production minimizes resource consumption and accelerates growth processes by culturing stem cells in controlled laboratory environments (Edelman et al., 2005). Moreover, cultured meat has the potential to be a

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https://doi.org/10.1016/j.jretconser.2025.104307
Received 12 November 2024; Received in revised form 27 March 2025; Accepted 2 April 2025
Available online 13 April 2025
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sustainable food production alternative (Fidder and Graça, 2023). A single muscle biopsy from a live cow could theoretically yield one billion hamburger patties within 45 days, a feat that requires 500,000 cows about 20 years via conventional methods (Hayflick, 1965). Cultured meat initiatives are not yet available to consumers in many countries, but they are gaining significant attention with support from prominent entrepreneurs and major agribusinesses (Durisin and Singh, 2018).

Cultured meats have garnered significant attention for their potential environmental, ethical, and political implications, but there remains a lack of understanding regarding consumer reactions to these innovative products. The service industry's success hinges on consumer acceptance, making it crucial to gauge public perception. Research into cultured meat, which is poised to replace traditional meat in the foreseeable future, is underway in various countries. Numerous studies are exploring consumer acceptance by focusing on variables, such as purchasing behavior, consumption patterns, and willingness to try cultured meat (Bryant and Barnett, 2020; van Dijk et al., 2023). Post et al. (2020) underscored that the cultured meat industry's capability to improve animal welfare and advance sustainable development relies heavily on its capacity to secure a significant portion of the mainstream market share. Many studies have concluded that consumers exhibit interest in cultured meat, and researchers have investigated acceptance levels, willingness to try, and willingness to pay for these types of products (Palmieri et al., 2021; Fu et al., 2023). However, research on cultured meat in China remains incomplete, and there is a limited amount of direct investigations into consumer awareness within the food service industry. It is important for ongoing research to address these gaps in understanding particularly as cultured meat continues to advance towards commercialization. Understanding consumer perceptions and behaviors will be instrumental in shaping the future trajectory of cultured meat adoption and integration into global food systems.

This study pursued two main objectives, which include investigating customer perceptions of cultured meat and identifying comprehending the factors that shape customer behavior towards cultured meat restaurants. The qualitative phase sought to explore factors that influence the preference for cultured meat via interviews and delving into the emotions and perceptions of the consumers by employing a mixed-methods approach. These factors were subsequently validated via a confirmatory factor analysis (CFA) using data from an online survey. The study also employed an importance-performance analysis (IPA) to pinpoint the aspects that the food service industry managers should prioritize based on the attributes of cultured meat. This paper aimed to bridge the gap between studies on cultured meat adoption and consumer perceptions in the food service industry by incorporating a fsQCA analysis and introducing a conceptual model.

2. Literature review

2.1. Cultured meat and its influence in the world

Cultured meat, also known as super meat and cultivated meat, refers to meat that is produced by an in vitro cell culture as opposed to being produced by typical animal farming methods. The cultured meat production process typically consists in five steps (see Fig. 1): cell isolation, cell culture, cell proliferation, tissue formation, and harvesting (Stephens et al., 2018; Tuomisto and Teixeira de Mattos, 2011). The process of cultured meat production involves several key steps. First, cell isolation extracts small samples of animal cells, such as muscle or stem cells from live animals without slaughtering them. These cells are then placed in a culture medium during cell culture, which provides essential nutrients for growth. Incubation in bioreactors and controlled environments that mimic internal animal conditions follow the cell culture. Next, during cell proliferation, the cells divide and multiply, which form muscle tissue with the addition of growth factors. The proliferated cells organize into muscle fibers in tissue formation, creating three-dimensional muscle tissue with adjustable properties. Harvesting occurs once the tissue reaches the desired size. The final step involves processing the harvested tissue and combining it with other ingredients, such as fat, protein, and flavorings to enhance the taste, texture, and the overall sensory experience. Cultured meat can replicate various traditional livestock products, including beef, pork, poultry, fish, and shrimp. Scientists can manipulate culture conditions to adjust characteristics, such as marbling, texture, and flavor to meet consumer preferences by aiming to replicate the sensory experience of traditional meat. From a retail perspective, cultured meat presents opportunities for high-end restaurants, boutique grocers, and premium food brands to market unique and sustainable alternatives to traditional meats (Specht et al., 2020). For instance, niche markets for environmentally conscious consumers or those seeking cruelty-free options are emerging, creating an avenue for early adopters to drive initial sales despite high production costs.

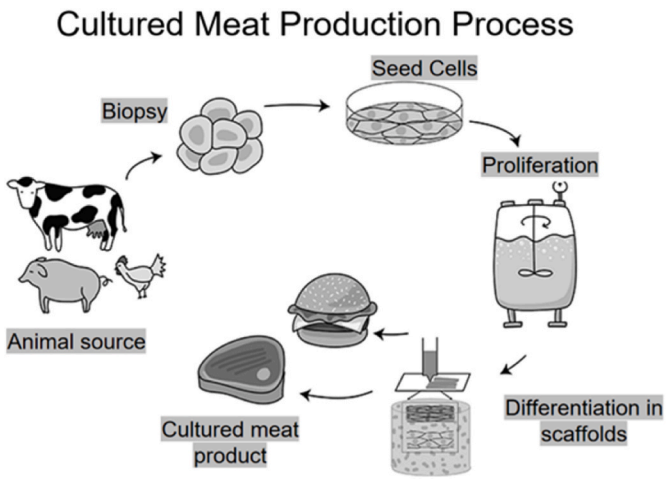


Fig. 1. Cultured meat production process (Image from Noble, 2023 and reconstructed by the author).

In 2020, Singapore was the first country to approve the sale of cultured meat products, and cultured meats are already commercially available. GOOD Meat, a seller of cultured meat, has been producing and selling chicken since December 2020 at special events in Singapore's luxury hotel restaurants (Amy, 2022). Moreover, some food service companies have already expressed interest in incorporating cultured meat into their menus. KFC announced plans in 2019 to trial a plant-based chicken product, which was developed in collaboration with Beyond Meat (Elizabeth Green, 2020). Pizza Hut Singapore more recently partnered with a cultured meat company to offer a limited-edition pizza that was topped with cultured chicken meat (Marc Cervera, 2022). Bryant et al. (2020) discovered that customers in China were positive about cultured meat as opposed to customers in the United States. Memphis Meats has produced cultured beef, chicken, and duck meat. Its investors include food giants Tyson Foods and Cargill. Shiok Meats, a Singapore-based company, produces cell-based seafood, including shrimp, crab, and lobster. Finless Foods, located in Silicon Valley, is also producing lab-grown seafood by focusing on tuna. The company already makes plant-based tuna. Mosa Meat, based in the Netherlands, is currently scaling up production of its lab-grown beef with what it calls the world's kindest burger (Fassler, 2021; Dale Cudmore, 2022; Buxton, 2022). Scaling up production and reducing production costs are the key challenges confronting the cultured meat industry beyond regulatory approval. Retailers face the challenge of balancing the high cost of cultured meat with consumer willingness to pay. Initial commercialization often targets premium segments to mitigate costs, but broader acceptance will require affordability and increased accessibility (Wilks et al., 2021; Ye et al., 2022). The current situation of cultured meat around the world overall is still in its early stages, but there is a growing amount of interest and investment in this

emerging technology (Kathryn, 2022). It is likely that more companies will enter the market as research and development efforts continue, and production costs will decrease, making cultured meat a more viable and accessible alternative to conventionally produced meat.

2.2. Cultured meat in China

China, the world's largest meat consumer, constitutes nearly 27 % of global meat consumption (McKinsey & Company, 2023). However, conventional meat production in the country has resulted in significant environmental repercussions (Bonny et al., 2015), prompting challenges, such as food safety issues, environmental degradation, and disease outbreaks within the meat industry. Cultured meat, cultivated from animal cells in laboratory conditions, offers a solution devoid of the need for traditional animal rearing and slaughtering (Kadim et al., 2015; Lee et al., 2022). This technology has seen recent advancements with several companies progressing towards commercialization, but Chinese cultured meat enterprises have encountered hurdles. High production costs and the technology's immaturity hinder large-scale production, which can potentially impact consumer acceptance (Bryant et al., 2020). Furthermore, uncertainty looms over the regulatory framework for cultured meat, and public perception remains mixed, with some individuals expressing safety and taste concerns (Wilks et al., 2021). The future trajectory of this technology in China remains intriguing given the ongoing research and regulatory clarity. Cultured meat presents significant opportunities for China by offering solutions for food security and environmental issues despite challenges by fostering innovation and technological growth as well as creating new avenues for business development. Cultured meat holds the potential to revolutionize China's meat industry and food system with governmental support and the endeavors of local companies.

2.3. Cultured meat characteristics and visit intention

Cultured meat production boasts significant resource efficiency by requiring notably less land, water, and energy while generating fewer emissions and waste products. Tuomisto et al. (2011) indicate that cultured meat production holds the potential to slash greenhouse gas emissions and land use by over 90 % compared to the traditional meat production. Another notable advantage of cultured meat is its potential to enhance animal welfare (Bryant et al., 2020). Cultured meat circumvents these ethical concerns by eliminating the need for raising and slaughtering animals, which is unlike traditional meat production that involves the breeding, confinement, and slaughtering of billions of animals annually. From a nutritional perspective, cultured meat is expected to be similar to conventional meat in terms of protein, fat, and nutrient content. However, because cultured meat is grown in controlled environments, its composition can be precisely adjusted and optimized for health benefits. This includes reducing harmful components such as saturated fats while increasing beneficial compounds like omega-3 fatty acids, which are essential for heart health (Post, 2014). For instance, Bryant et al. (2020) found that cultured chicken meat contained a lower fat content and a higher protein concentration compared to traditionally produced chicken meat. Furthermore, cultured meat eliminates the risk of antibiotic resistance and zoonotic diseases, which are significant concerns in conventional livestock farming (Ramani et al., 2021). Additionally, cultured meat also has the potential to address the issue of food security particularly in areas where access to traditional meat is limited (Wikandari et al., 2021). Despite these benefits, cultured meat raises concerns related to potential health risks, such as allergens, as well as its safety and quality (Rombach et al., 2022). Additionally, the high cost of production remains a significant challenge, as it currently exceeds the production costs of conventional meat. This is primarily due to the expensive growth media required for cell cultivation, the high costs of bioreactors and other specialized equipment, and the immature supply chains for necessary materials such as scaffolds and growth factors (Specht et al., 2020; Ye et al., 2022). Furthermore, the lack of economies of scale and the significant investment in research and development further drive up costs (Post, 2012). However, ongoing research and development efforts are expected to drive down production costs over time which enhances the accessibility for consumers. Consumer attitudes toward novel food technologies, which include irradiation and genetic modification, significantly influence purchasing decisions (Cardello, 2003; Saini et al., 2024; Yong et al., 2024). Trust in information sources, risk perceptions, and socio-demographic factors also play a role in acceptance (Costa-Font et al., 2008; Klerck and Sweeney, 2007; Lee et al., 2024; Yu et al., 2024). Some consumers are hesitant to embrace cultured meat due to the taste, safety concerns, and perceptions of unnaturalness despite interest in novel foods (Evans and Johnson, 2021).

Behavior intention, which is defined by Zeithaml et al. (1996) and Akram et al. (2024), signifies an individual's readiness to engage in a specific behavior. Duan et al. (2020) emphasize its subjective likelihood of participating in a particular activity. Food attributes wield considerable influence on behavior intention especially concerning novel foods (Jung et al., 2022). Novel foods, which are unfamiliar to specific cultures or populations, are heavily influenced by attributes, such as taste, texture, appearance, and perceived healthiness, which impact behavior intention. Taste, a critical factor in food acceptance, varies among individuals, and it influences their likelihood of visiting a restaurant that serves cultured meat if satisfied with its taste and texture. Appearance similarly influences food acceptance, and individuals have varying preferences for the food's appearance (Razzaq et al., 2021). Perceived healthiness also plays a pivotal role, and several studies suggest that cultured meat may offer a more favorable nutritional profile than conventionally produced meat, which is potentially appealing to health-conscious consumers (Liu and Kim, 2024). Finally, price is a significant consideration for consumers (Raza et al., 2024), and the current higher production costs of cultured meat compared to conventional meat potentially deter price-sensitive consumers from visiting restaurants that offer cultured meat dishes (Specht et al., 2020). We can therefore assume that various attributes of cultured meat, which include perceived sustainability, healthiness, taste, texture, and price, are likely to shape consumers intentions to visit restaurants that offer these types of products.

3. Methodology

3.1. Study process and design: a mixed methods approach

The current study consists of two studies (see Fig. 2), which analyze the factors that affect exploring for new cultured meat attributes and the intention to visit cultured meat restaurants. First, the integration of the qualitative and quantitative methods is expected to be used as a reasonable research method to accurately identify the perceptions of cultured meat according to the consumer. Second, an IPA analysis (Martilla et al., 1977) was additionally used to evaluate the importance of the different factors for the customers and to prioritize the factor performance in improving the factor performance in the cultured meat industry. The IPA assesses the perceived importance and performance of various service attributes of the consumers, which categorizes them into four quadrants (Xu et al., 2022). These quadrants aid in pinpointing areas for improvement and prioritizing actions in enhancing customer satisfaction.

Structural equation model (SEM) analysis is also conducted to determine the net effects between the attributes and behavior intention, whereas fsQCA is employed to identify the most suitable set of factors that influence the intention to visit in relation to cultured meat attributes. Increasingly utilized in the food service industry, hospitality, and tourism management, fsQCA analysis aims to identify the configurations of conditions that lead to specific outcomes (Rasoolimanesh et al., 2021; Wang et al., 2021; Alshehri, 2025; Sharkasi et al., 2025). In addition, the

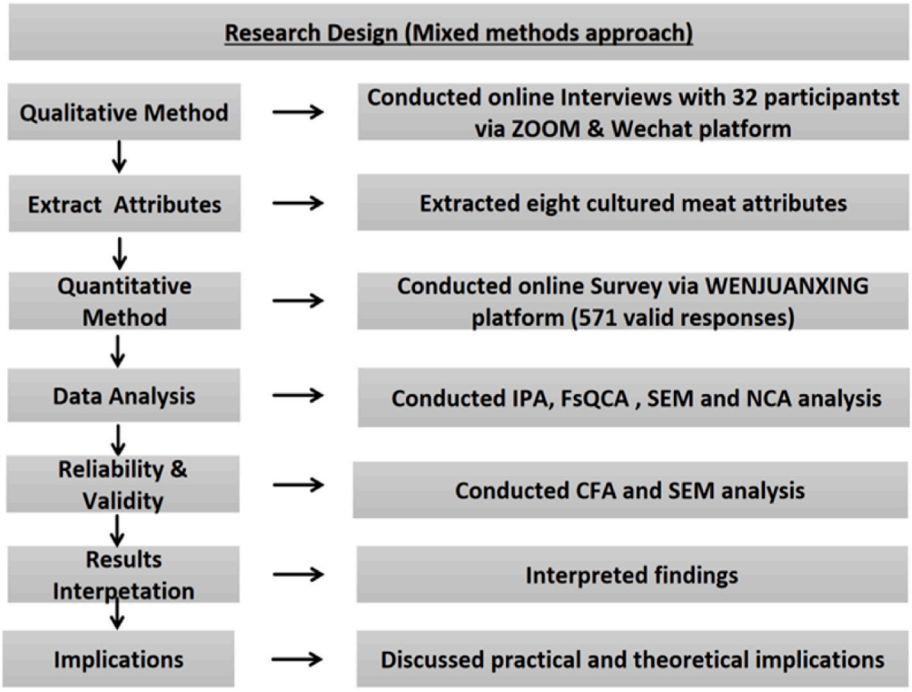


Fig. 2. Research design process.

fsQCA method is widely used in research on plant-based meat, organic food, and sausage meat, although it is not cultured meat (Bhattacharyya et al., 2023; Han and Wu, 2021). The fsQCA is a set-theoretic method that enables the analysis of multiple necessary and sufficient conditions for particular outcomes (De Vos and Cambré, 2017). It proves valuable in scenarios with multiple pathways and nonlinear or noncausal relationships between variables. FsQCA is apt for analyzing conditions that lead to various outcomes, such as customer satisfaction, loyalty, and intention to return in the food service industry, hospitality, and tourism management contexts (Xu et al., 2022). Numerous studies have employed fsQCA to investigate conditions that yield positive outcomes in these fields.

The combination of IPA and fsQCA was selected to provide a comprehensive understanding of consumer behavior. IPA highlights areas requiring improvement, fsQCA explores complex configurations of factors. These methods were chosen over others for their complementary strengths and ability to address the multidimensional nature of the study's objectives.

3.2. *Qualitative approach and quantitative approach*

Online interviews were conducted with 32 participants as part of the qualitative data collection process. The sample size was determined based on data saturation rather than statistical representation, a common practice in qualitative research (Guest et al., 2006). Hennink et al. (2017) suggest that data saturation typically occurs within 20–30 interviews, depending on the study design and topic complexity. To strengthen the reliability of the findings, additional measures were taken to ensure the trustworthiness of the qualitative data. To ensure credibility, interviewees were selected from diverse groups within the food service and hospitality sectors, incorporating a range of perspectives. This included food service industry workers and non-food service industry workers, food service industry experts, operators, restaurant managers, official workers, hospitality workers, as well as master's and doctoral degree holders in hospitality and tourism management, and meat lovers. The goal was to better understand the indicators customers use to evaluate cultured meat consumption. Member checks were also incorporated, allowing participants to verify or clarify their responses. For qualitative data analysis, an inductive approach was used to allow themes and patterns to emerge directly from the data, ensuring the findings reflected the participants' perspectives. The participants freely shared their opinions on the attributes of cultured meat. Prior to the interviews, they were provided with detailed information, including text and video, about cultured meat. The interviews focused on several key questions: 1) How would you describe the attributes and key advantages of cultured meat? 2) What attributes motivate you to try cultured meat? 3) What attributes prevent you from trying cultured meat? 4) What are your biggest concerns about trying cultured meat?

This study collected data using a web-based platform provided by Wenjuanxing (www.wjx.cn). This data collection method is considered reliable as it enhances research validity through randomization. Systematic random sampling was employed to ensure the survey was evenly distributed to respondents across various regions in China, thereby minimizing selection bias. To improve the clarity of the sampling approach and address potential concerns regarding representativeness, the initial sample frame was designed to ensure diversity across key demographic groups, including age, gender, education level, and geographic distribution. The study focused on Chinese consumers, selecting a sample with significant representation of meat consumption to maximize the generalizability of the findings. However, we acknowledge the limitations in generalizing the results to other cultural contexts.

To address common method bias, the survey was conducted online, and the system was organized to allow only one response per IP address. Additionally, prior to the interviews and survey, detailed explanations were provided to ensure the anonymity of respondents. The respondents for the qualitative and quantitative studies were also set up to avoid overlap. Furthermore, a pre-test survey was distributed to 30 participants before the full survey to identify and correct any ambiguous terms or difficult-to-understand sections. This process improved the clarity of the survey and ensured that respondents could better understand the questions. To reduce non-response bias, the survey was set up to prevent submission unless it was fully completed.

3.3. Conceptual model development

The research draws on an online interview to extract eight attributes that are related to cultured meat. The study introduces an innovative methodological approach by employing both symmetrical and non-symmetrical paradigms. The proposed model encompasses attributes, such as cultured meat safety, quality, nutrients, sustainability, animal welfare, price, novelty, and aversion. The outcome variable is the intention to visit a cultured meat restaurant. The previous literature in the food service industry highlights attributes that directly influence visit intention (Kunz et al., 2021). Associations between these variables were explored within the context of cultured meat by building upon the existing studies. Figs. 3 and 4 illustrate the proposed conceptual model. The questionnaire, which was evaluated using a 7-point Likert scale, aims to gauge participant agreement. The following hypotheses are formulated accordingly.

Proposition 1(a-h). *The attributes of cultured meat significantly influence intention to visit a cultured meat restaurant.*

Proposition 2. *The attributes of cultured meat have an optimum combination influence on intention to visit a cultured meat restaurant.*

4. Results

4.1. Interview profile of the respondents

Chinese interviewees, all in their 20s–40s, were interviewed via the ZOOM and Wechat platforms from January 14 to 28, 2023 to discover the attributes of cultured meat. Each interview lasted about 30–40 min, and the participants were free to ask, discuss, and answer questions. Most of the interviewers accounted for 20 people in their 30s. The interviewees were provided detailed information, images, and questions related to cultured meat before the interview. To ensure that the qualitative data reached saturation, the interviews were conducted until no new themes or attributes emerged. Saturation was observed by the 29th interview, where repeated patterns and insights about cultured meat attributes became evident. The additional three interviews confirmed these patterns and did not contribute any new information, validating that the sampling size was sufficient for exploratory qualitative research. The following is a summary of the key findings from the statements of the 32 interviewees.

> "… I think the biggest advantage of cultured meat is environmental protection because it doesn't require a lot of land and space to produce cultured meat. I think it is a good technology that can positively change the agricultural system."
>
> **-No. 1 and 7**

> "… Cultured meat is not familiar to me. I am a meat lover. If I could enjoy meat without animal slaughtering, I would gladly choose cultured meat."
>
> **-No. 12 and 22**

> "… Whenever I eat meat, I don't think too much about killing animals because I regard those pigs, cows, and chickens as an indispensable part of human food. However, it would be better if we can solve human food without killing animals."
>
> ***-No. 2, 9, 19, and 28***

> "… it is not easy to be infected with viruses and diseases, such as the avian influenza and Covid-19. Cultured meat is also environmentally friendly and rarely comes into conflict with nature. Cultured meat is new, and it is an opportunity to try something different for me."
>
> **-No. 29 and 32**

It is noted that some people are excited about the potential benefits of cultured meat. For example, they see it as a way to address environmental concerns that are associated with traditional animal agriculture, such as greenhouse gas emissions, water use, and diseases transmitted from livestock. Some individuals also believe that cultured meat could be a more ethical and humane way to produce meat, since it does not require the slaughter of animals. Most respondents were satisfied with the benefits of producing cultured meat, which were even admired by individuals who said they would never try cultured meat.

> "… The most worrisome thing is that the chewiness of cultured meat cannot catch up with traditional meat, and the appearance is also



Fig. 3. Model for sufficient predictors: The net effect.

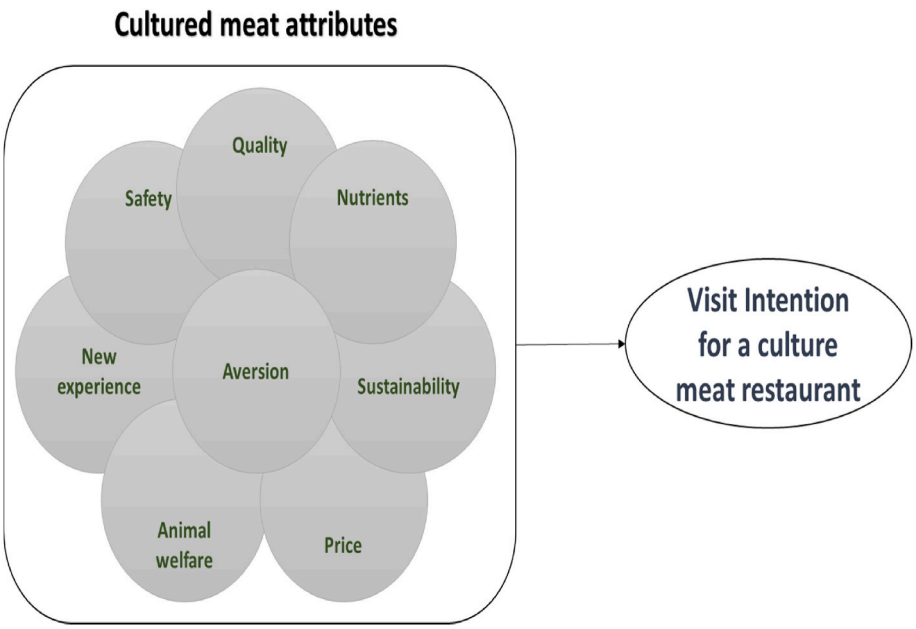


Fig. 4. The research configurational model.

very important to me. If the texture or taste isn't the same, I'd probably choose traditional meat despite the animal sacrifices"

-No. 9, 26, and 31

"… Looking at the cultured meat information, I think it is good for health. This is because the related substances provided by cultured meat can be richer and more nutritious than conventional meat, such as fat, protein, and various nutrients can be controlled.

-No. 29 and 32

According to these discussions, the respondents considered the quality of cultured meat. They showed a lot of interest in the texture, color, flavor, and nutrients of cultured meat. Visual aspects of cultured meat were discussed by over half of the respondents.

"… I still feel that cultured meat is unnatural and a bit disgusting. I have tried plant-based meat before, and I have no hesitation in regard to trying it again. However, cultured meat is different. Cultured meat feels like it's made entirely from chemicals. The most important thing is my health problem."

-No. 3, 4, 20, 21, 26, 27, and 28

"… many people think that cultured meat is not real meat, and it is not natural, which includes me. If you don't tell me it's cultured meat I might try it, but once I know it's cultured meat, I won't eat it at least now."

-No. 13, 14, 18, and 27

On the other hand, some people are hesitant to eat lab-raised meat for the reasons discussed above. They may be concerned about the safety of the technology or the nutritional quality of the meat.

"… Cultured meat is very attractive to me, and I am also curious whether it tastes the same as traditional meat. If there is a cultured meat restaurant in my hometown, I want to immediately try it. I like to try different foods. I hope it's not too expensive."

-No. 10,11,15, and 16

" …. The price of cultured meat is a major concern for me. I watched a YouTube video that is related to lab-raised meat. A small piece of chicken was too expensive. If I can try cultured meat with a reasonable price, I want to try. I will bring all my friends and family."

-No. 5, 8, and 30

We can see from these discussions that some respondents are interested in cultured meat. However, these individuals are concerned about the price of cultured meat. Attitudes towards cultured meat are susceptible to being influenced by a variety of variables, including individual beliefs and values, cultural and social norms, and the perceptions of the technology's safety and effectiveness. It will be interesting to see how public opinion about cultured meat changes as the technology continues to evolve and become more widespread.

A total of 26 items were extracted based on the views and thoughts of the interviewees. The 26 items include animal diseases, antibiotic misuse, epidemics, taste, aroma, appearance, fat control, protein control, nutrient control, reduce land/water/energy use, reduce greenhouse gas emissions, eco-friendliness, reduce intensive livestock production, reduce livestock slaughtering, animal friendly, price reasonableness, appropriateness, affordablity, food curiosity, new experience, trying different foods, unnaturalness, food safety, healthiness, and food neophobia. These variables were modified according to the previous literature to conduct the survey (Zhang et al., 2020; Tomiyama et al., 2020; Fraeye et al., 2020; Nobre, 2022; Hamdan et al., 2021).

4.2. *Profile of the survey respondents*

The survey questionnaire was prepared based on the interview results and was distributed in mainland China from February 4 to 28, 2023. The survey was conducted in Chinese to ensure that participants fully understood the questions and could provide accurate responses. The first part of the survey describes the purpose of the survey, and it provides the email address and phone number for the inquiries that are related to the survey. The survey participants must be over the age of 20, each participant cannot repeatedly submit the questionnaire, and all the questions must be filled out before submission. In addition, the inquiry commenced following a comprehensive explanation about the concept and production process of cultured meat given the anticipated lack of familiarity with cultured meat. A total of 614 responses were collected, with 571 valid samples retained for analysis after the removal of outliers. The response rate for the survey was 93.0 %. Outliers were identified based on extreme response patterns, such as unrealistically short completion times (e.g., under 3 min) or logical inconsistencies in responses. These criteria were established to ensure the integrity and robustness of the data. 301 (52.7 %) of the total respondents were male,

and 270 (47.3) of them were female. The 20 to 39 age group comprised of the largest proportion at 56.7 %, which was followed by individual's in the 50 to 59 age group at 18.7 % and individuals in the 40 to 49 age group at 18 %. Regarding education, 12.1 % of the respondents held a high school diploma, 29.8 % had completed college with 2 or 3 year degrees, 44.1 % possessed a bachelor's degree, and 14 % held a master's or doctoral degree. In terms of income, 22.1 % of the respondents earned between 50,001 and 100,000 yuan, which is approximately equivalent to $1468, 21.4 % earned between 100,001 and 150,000 yuan, 12.8 % earned between 150,001 and 200,000 yuan, 10.3 % earned over 200,000 yuan, and 27.7 % earned less than 50,000 yuan. Lastly, most respondents, which accounted for 90.5 % (n = 517), had no experience with cultured meat (Table 1).

4.3. Reliability and validity outcomes

Exploratory factor analysis (EFA), utilizing principal components analysis (PCA) and a varimax rotation, was conducted to reveal the fundamental dimensions of cultured meat attributes, as displayed in Table 1. The analysis identified 26 items that are distributed across 8 factors. These factors encompassed 3 items each for safety, including animal disease, antibiotic misuse, and epidemics; cultured meat quality, including include taste, aroma, and appearance; nutrients, including fat control, protein control, and nutrients control; sustainability, including reduced land/water/energy use, reduced greenhouse gas emissions, and eco-friendliness; animal welfare, including reduce intensive livestock production, reduce livestock slaughtering, and animal friendliness; price, including reasonableness, appropriate, and inexpensive; new experience, including food curiosity, new experience, and trying different foods; and aversion to cultured meat, including unnaturalness, food safety, healthiness, and food neophobia. The appropriateness of the EFA was validated with a Kaiser-Meyer-Olkin value of 0.745 and a statistically significant Bartlett's test of sphericity ($p < 0.001$) (George and Mallery, 2001). Eigenvalues greater than 1 were observed for the 8 primary dimensions, which collectively explain 80.038 % of the total variance. These dimensions were labeled safety of consuming cultured meat (SCM), quality of cultured meat (QCM), nutrients of cultured meat (NCM), sustainability (SUST), animal welfare (AW), price of cultured meat (PCM), new experience (NE), and aversion to cultured meat (ACM).

Table 2 illustrates that three items explain the variance of 9.408 % in SCM, four items of the QCM explain the variance of 10.740 %, and three items of the NCM accounted for 9.736 %. Three items of SUST, AW, PCM, NE, and four items of ACM explained the variance of 9.515 %, 9.690 %, 9.815 %, 9.049 %, and 12.085 %, respectively. In addition, eight factors were larger than Cronbach's alpha coefficient of 0.80 in the reliability analysis, so the reliability validity was confirmed.

4.4. Importance-performance measurement

The mean calculations were performed for the importance and performance of the 26 cultured meat attributes to maintain consistency with the IPA framework, which are shown in Table 3. The use of importance-performance analysis (IPA) effectively identifies priority areas for improvement and potential overinvestment by categorizing attributes into four quadrants based on their importance and performance (Martilla and James, 1977). The first quadrant is ***keep up the good work***, which contains two variables, including provide controlled fat and inexpensive. The second quadrant ***is possible overkill***, which contains eleven variables, including provide controlled protein, provide controlled nutrients, less use of land, water, and energy, reduction of greenhouse gas emissions, eco-friendliness, reduction of livestock production, reduction of livestock slaughtering, animal-friendliness,

Table 1
Survey profile information (N571).

Variable	Category	Number	%
Gender	Male	301	52.7 %
	Female	270	47.3 %
Age	20~29	161	28.2 %
	30~39	163	28.5 %
	40~49	103	18 %
	50~59	107	18.7 %
	60 years old and above	37	6.5 %
Education	High school or below	69	12.1 %
	Vocational Technical college education	170	29.8 %
	Bachelor's degree	252	44.1 %
	Master's degree	44	7.7 %
	Doctorate or above	36	6.3 %
Occupation	Institutional organization/Civil servant	218	38.2 %
	Company staff	178	31.2 %
	Full-time self-employed	79	13.8 %
	Student	49	8.6 %
	Other	47	8.2 %
Annual Income (RMB)	Below 50,000	158	27.7 %
	50,000–100,000	126	22.1 %
	100,001–150,000	122	21.4 %
	150,001–200,000	73	12.8 %
	200,001–300,000	59	10.3 %
	Above 300,000	33	5.8 %
Cultured meat experience	Yes	54	9.5 %
	No	517	90.5 %

Table 2
EFA.

Scale item	Loadings	Eigenvalues	Variance explained	Cronbach's alpha
Safety of cultured meat				
SCM1	0.878	2.108	9.408	0.882
SCM2	0.903			
SCM3	0.900			
Quality of cultured meat				
QCM1	0.777	3.259	10.740	0.852
QCM2	0.851			
QCM3	0.884			
QCM4	0.802			
Nutrients of cultured meat				
NCM1	0.890	2.620	9.736	0.905
NCM2	0.920			
NCM3	0.918			
Sustainability				
SUST1	0.887	2.243	9.515	0.890
SUST2	0.906			
SUST3	0.902			
Animal welfare				
AW1	0.874	2.376	9.690	0.900
AW2	0.935			
AW3	0.920			
Price of cultured meat				
PCM1	0.902	2.713	9.815	0.903
PCM2	0.921			
PCM3	0.912			
New experience				
NE1	0.812	1.919	9.049	0.857
NE2	0.919			
NE3	0.913			
Aversion to cultured meat				
ACM1	0.864	3.572	12.085	0.907
ACM2	0.909			
ACM3	0.912			
ACM4	0.842			
Total experience			80.038	

Kaiser-Meyer-Olkin (KMO) = 0.745.
Note: SCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.

Table 3
IPA of cultured meat attributes.

No.	Items	Measurement item	Imp (M)	Per (M)	I-P (Q)	t-value	p-value
1	SCM1	Prevents diseases	4.81	4.11	3	6.324	0.000
2	SCM2	Less use antibiotics	4.81	4.05	3	7.069	0.000
3	SCM3	Prevents infectious diseases	4.91	4.06	4	7.323	0.000
4	QCM1	Taste/flavor	5.05	3.96	4	11.840	0.000
5	QCM2	Chewing texture	5.04	4.04	4	10.450	0.000
6	QCM3	Aroma	5.2	4.17	4	10.604	0.000
7	QCM4	Appearance (color)	5.34	4.18	4	12.216	0.000
8	NCM1	Provide controlled fat	4.9	4.43	1	4.374	0.000
9	NCM2	Provide controlled protein	4.79	4.59	2	1.909	0.057
10	NCM3	Provide controlled nutrients	4.84	4.6	2	2.123	0.034
11	SUST1	Less use land/ water/energy	4.77	4.48	2	2.557	0.011
12	SUST2	Reduces greenhouse gas	4.76	4.43	2	2.881	0.004
13	SUST3	Eco-friendly	4.79	4.61	2	1.465	0.143
14	AW1	Reduces livestock production	4.74	4.73	2	0.031	0.976
15	AW2	Reduces livestock slaughter	4.75	4.83	2	−0.655	0.513
16	AW3	Animal-friendly	4.71	4.75	2	−0.382	0.702
17	PCM1	Reasonable price	4.68	4.41	2	2.332	0.020
18	PCM2	Appropriate price	4.7	4.41	2	2.607	0.009
19	PCM3	Inexpensive	4.86	4.52	1	2.905	0.004
20	NE1	Curiosity	4.72	4.3	3	3.445	0.001
21	NE2	New experience	4.59	4.31	3	2.389	0.017
22	NE3	Taste different foods	4.69	4.55	2	1.162	0.246
23	ACM1	unnatural	4.88	3.77	4	11.335	0.000
24	ACM2	Concern about food safety	4.86	3.83	4	10.100	0.000
25	ACM3	Concern about health safety	4.93	4.09	4	8.705	0.000
26	ACM4	Food neophobia	4.94	4.13	4	8.706	0.000

Note: Imp: important, Per: Performance, SCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.

reasonable price, appropriate price, and taste different food. Curiosity, new experience, prevents diseases, and less use antibiotics were in the ***low priority*** quadrant, which contains four variables. Nine variables appeared in the ***concentrate here quadrant***, which included prevents infectious diseases, taste, flavor, chewing texture, aroma, appearance (color), unnatural, concern about food safety, concern about health safety, and food neophobia. The results indicate that businesses should focus their efforts on improving the attributes in the "Concentrate Here" quadrant, as these are areas where improvement can have the most significant impact on consumer perceptions and purchasing behavior. Specifically, sensory qualities like taste, texture, and aroma, along with improving consumer confidence in food safety, should be prioritized.

4.5. Confirmatory factor analysis (CFA)

Confirmatory factor analysis (CFA) was conducted to assess convergence and discriminant validity among the study constructs, as well as to evaluate internal consistency and reliability. A set of 26 items was utilized to measure the cultured meat attributes. The goodness-of-fit statistics of the measurement model (χ 2 = 839.719, df = 396, GFI = 0.916, CFI = 0.959, TLI = 0.952, and RMSEA = 0.044) indicated a satisfactory fit of the conceptual model. These values are considered reliable because they meet or surpass widely accepted thresholds in the literature: the GFI above 0.90 suggests an adequate fit (Hair et al., 2019); the CFI and TLI above 0.95 indicate excellent fit (Hu and Bentler, 1999); and the RMSEA below 0.05 represents a close fit to the population covariance matrix (Browne and Cudeck, 1992). All factor loadings of the items measuring the cultured meat attributes exceeded 0.5. The composite reliability (CR) values for each study construct surpassed the recommended threshold of 0.60 (Hair et al., 2019), which ranged from 0.855 to 0.923. The average variance extract (AVE) scores for all study variables also exceeded the standard criterion of 0.50, which indicates appropriate convergent validity based on Fornell and Larcker, (1981) guidelines. Discriminant validity was assessed by comparing the AVE values with the squared correlations between variables, and it met acceptable levels. The detailed CFA results are presented in Tables 4 and 5.

4.6. Sufficient antecedents

Structural equation modeling (SEM) was employed to investigate the net effects. The goodness-of-fit statistics (χ 2 = 842.995, df = 398, GFI = 0.916, CFI = 0.959, TLI = 0.952, and RMSEA = 0.044) revealed an appropriate results. Visit intention for a cultured meat restaurant (VICMR) was found to be influenced by safety of consuming cultured meat (SCM: $\beta = 0.093$ and $p < 0.05$), quality of cultured meat (QCM: $\beta =$

Table 4
CFA.

Scale item	β	Mean	S.D
Safety of cultured meat			
SCM1	0.807	4.81	1.838
SCM2	0.881	4.81	1.747
SCM3	0.85	4.91	1.888
Quality of cultured meat			
QCM1	0.687	5.05	1.442
QCM2	0.785	5.04	1.445
QCM3	0.874	5.20	1.440
QCM4	0.733	5.34	1.415
Nutrients of cultured meat			
NCM1	0.824	4.90	1.859
NCM2	0.888	4.79	1.867
NCM3	0.904	4.84	1.900
Sustainability			
SUST1	0.836	4.77	1.823
SUST2	0.873	4.76	1.852
SUST3	0.854	4.79	1.908
Animal welfare			
AW1	0.777	4.74	1.903
AW2	0.94	4.75	1.895
AW3	0.886	4.71	1.994
Price of cultured meat			
PCM1	0.85	4.68	1.854
PCM2	0.901	4.70	1.827
PCM3	0.862	4.86	1.947
New experience			
NE1	0.664	4.72	1.962
NE2	0.907	4.59	1.846
NE3	0.893	4.69	1.969
Aversion to cultured meat			
ACM1	0.812	4.88	1.534
ACM2	0.886	4.86	1.572
ACM3	0.889	4.93	1.629
ACM4	0.785	4.94	1.607
Visit intention for a cultured meat restaurant			
VICMR1	0.739	5.03	1.451
VICMR2	0.784	5.01	1.419
VICMR3	0.843	4.90	1.486
VICMR4	0.847	5.03	1.444
VICMR5	0.803	5.01	1.438

Note: VICMR: visit intention for a cultured meat restaurant, SCCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.

Table 5
Results of measurement model and correlations.

	ACM	SCM	QCM	NCM	SUST	AW	PCM	NE	VICMR
AW	0.844								
SCM	−0.102	**0.895**							
QCM	0.053	0.117	**0.773**						
NCM	0.023	0.146	0.034	**0.873**					
SUST	0.090	0.046	0.136	0.175	**0.854**				
AW	0.033	0.040	0.060	0.071	0.032	**0.870**			
PCM	−0.056	0.038	0.055	0.018	0.020	0.147	**0.861**		
NE	0.051	0.026	−0.019	0.045	−0.028	0.022	0.016	**0.829**	
VICMR	−0.123	0.151	0.151	0.151	0.130	0.142	0.219	−0.007	**0.804**
CR	0.908	0.923	0.855	0.905	0.890	0.903	0.896	0.866	0.901
AVE	0.712	0.801	0.598	0.762	0.730	0.757	0.741	0.687	0.647
MSV	0.015	0.023	0.023	0.031	0.031	0.022	0.048	0.003	0.048

Goodness-of-Fit Statistics: χ 2 = 839.719, df = 396, GFI = 0.916, CFI = 0.959, TLI = 0.952, RMSEA = 0.044.
Note: VICMR: visit intention for a cultured meat restaurant, SCCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.

0.111 and $p < 0.05$), nutrients of cultured meat (NCM: $\beta = 0.107$ and $p < 0.05$), sustainability (SUST: $\beta = 0.097$ and $p < 0.05$), animal welfare (AW: $\beta = 0.098$ and $p < 0.05$), price of cultured meat (PCM: $\beta = 0.184$ and $p < 0.001$), and aversion to cultured meat (ACM: $\beta = -0.125$ and $p < 0.01$). The factor of new experience (NE: $\beta = -0.008$ and $p > 0.05$) did not yield a significant effect on VICMR. The structural model fit the data, which was based on the goodness-of-fit statistics. The findings from the SEM analysis align with previous studies that examined the impact of restaurant attributes, food choice factors, and characteristics of novel food on consumer behavior (Liu and Tse, 2018; Jung et al., 2022). Table 6 illustrate that all factors except for new experience significantly influence the visit intention for a cultured meat restaurant. The factor of aversion to culture meat has a negative effect on the visit intention. Food neophobia has a negative impact on the on-purchase intent, attitudes toward novel foods, and willingness to try, which is similar to a previous study (Baker et al., 2016). Previous research also found that safety of health positively influences customer behavior (Verbeke et al. (2015). Moreover, the price of cultured meat has a positive effect on visit intention, which is inconsistent with a previous study. A study by Jung et al. (2022) found that the higher the price of a novel food, the higher the purchase intention of consumers for the new food. The research also shows that food quality and food nutrients influence customer attitudes and purchase intentions, which is similar to the findings by Rana and Paul, (2017). Furthermore, this study highlighted the influence of cultured meat production sustainability on the intention to visit a cultured meat restaurant, which is consistent with the findings by Han (2021). The factor of animal welfare emerged as crucial in cultured meat research, which exhibited a positive influence on visit intention in this study and significantly contributed to behavioral intention in the prior research (Siddiqui et al., 2022). However, the factor of new experience showed no effect on visit intention, which contradicted the findings from previous studies on insect-related food and behavioral intention (Stone et al., 2022). This confirms that people are more concerned with other factors than curiosity when they choose food that is made from cultured meat. All of these results therefore support the establishment of Proposition 1. The SEM results, displayed in Table 6, reveal the net effect of cultured meat attributes on the outcomes of visit intention, but the results that require predicting behavioral intentions by calculating heterogeneous relationships between the antecedent factors and combinations of the antecedent factors, such as causal recipe should be calculated.

Table 6
SEM.

	Hypothesis →	Coefficients	t-value	Status	
H1a	SCM	VICMR	0.093	2.026*	S
H1b	QCM	VICMR	0.111	2.408*	S
H1c	NCM	VICMR	0.107	2.349*	S
H1d	SUST	VICMR	0.097	2.119*	S
H1e	AW	VICMR	0.098	2.221*	S
H1f	PCM	VICMR	0.184	4.071***	S
H1g	NE	VICMR	−0.008	−0.181	NS
H1h	ACM	VICMR	−0.125	−2.806**	S

Goodness-of-Fit Statistics: χ 2 = 842.995, df = 398, GFI = 0.916, CFI = 0.959, TLI = 0.952, RMSEA = 0.044.
Note: VICMR: visit intention for a cultured meat restaurant, SCCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.
*P < 0.05, **P < 0.01, ***P < 0.001.

4.7. *Fuzzy-set qualitative comparative analysis – sufficient configurations*

This experiential study also investigated the sufficient configurations to describe the causal recipes that caused high or low visit intention for a cultured meat restaurant. In asymmetric modeling analysis, coverage and consistency thresholds are typically set at 0.20 and 0.80, respectively (Ragin, 2009). Tables 7 and 8 illustrate seven causal conditions that involve Model 1, which includes a high level of SCM, NCM, SUST, and PCM and a low level of AW, NE, and ACM; Model 2, which includes a high level of QCM, NCM, SUST, and PCM and a low level of AW, NE, and ACM; Model 3, which includes a high level of SCM, NCM, PCM, and NE and a low level of SUST, AW, and ACM; Model 4, which includes a high level of QCM, AW, PCM, and NE and a low level of SCM, SUST, and ACM; Model 5, which includes a high level of NCM, AW PCM, and NE, and a low level of SCM, SUST, and ACM; Model 6, which includes a high level of SCM, QCM, NCM, SUST, and NE and a low level of AW and ACM; and Model 7, which includes a high level of SUST, PCM, and NE and a low level of SCM, QCM, NCM, AW, and ACM, facilitated the high visit intention for a cultured meat restaurant (coverage: 0.364 and consistency: 0.966). On the other hand, low visit intention for a cultured meat restaurant was caused by five causal recipes (coverage: 0.337 and consistency: 0.932) that involved Model 1, which had a high SCM level and a low level for QCM, NCM, SUST, AW, PCM, and NE; Model 2. Which had a high PCM level and a low level for SCM, QCM, NCM, SUST AW, and NE; Model 3, which had a high NE level and a low level for SCM, ACM, QCM, NCM, SUST, and AW; Model 4, which had a high level for PCM, NE, and ACM and a low level for QCM, NCM, SUST, and AW; and Model 5, which had a high level for SCM, SUST, PCM, and ACM and a low level for QCM, NCM, AW, and NE). These types of configurations revealed that the factors have an optimum combination influence on visit intention for a cultured meat restaurant, which supports the establishment of Proposition 2.

4.8. *Necessary conditions*

Table 9 shows the results of the necessary condition analysis, which

Table 7
Causal models for high visit intention for a cultured meat restaurant.

A. VICMR = f (SCM, QCM, NCM, SUST, AW, PCM, NE, ACM)	Raw Coverage	Unique Coverage	Consistency
Model 1: SCM*NCM*SUST*~AW*PCM*~NE*~ACM	0.2119	0.0052	0.9810
Model 2: QCM*NCM*SUST*~AW*PCM*~NE*~ACM	0.2121	0.0055	0.9794
Model 3: SCM*NCM*~SUST*~AW*PCM*NE*~ACM	0.2011	0.0135	0.9824
Model 4: ~SCM*QCM*~SUST*AW*PCM*NE*~ACM	0.2126	0.0110	0.9772
Model 5: ~SCM*NCM*~SUST*AW*PCM*NE*~ACM	0.2016	0.0047	0.9808
Model 6: SCM*QCM*NCM*SUST*~AW*NE*~ACM	0.2318	0.0317	0.9849
Model 7: ~SCM*~QCM*~NCM*SUST*~AW*PCM*NE*~ACM	0.1800	0.0075	0.9833
Solution Coverage: 0.3642			
Solution Consistency: 0.9655			
Causal models for low visit intention for a cultured meat restaurant			
~A. ~VICMR = f (SCM, QCM, NCM, SUST, AW, PCM, NE, ACM)	Raw Coverage	Unique Coverage	Consistency
Model 1: SCM*~QCM*~NCM*~SUST*~AW*~PCM*~NE	0.2280	0.0139	0.9552
Model 2: ~SCM*~QCM*~NCM*~SUST*~AW*PCM*~NE	0.2275	0.0096	0.9564
Model 3: ~SCM*~QCM*~NCM*~SUST*~AW*NE*ACM	0.2549	0.0077	0.9541
Model 4: ~QCM*~NCM*~SUST*~AW*PCM*NE*ACM	0.2537	0.0080	0.9535
Model 5: SCM*~QCM*~NCM*SUST*~AW*PCM*~NE*ACM	0.2265	0.0256	0.9588
Solution Coverage: 0.33696			
Solution Consistency: 0.931734			

Note: VICMR: visit intention for a cultured meat restaurant, SCCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.

Table 8
Configurations for high/low level of visit intention for a cultured meat restaurant.

	High level of VICMR							Low level of VICMR				
Configuration	M1	M2	M3	M4	M5	M6	M7	M1	M2	M3	M4	M5
SCM	●		●	⊗	⊗	●	⊗	●	⊗	⊗		●
QCM		●		●		●	⊗	⊗	⊗	⊗	⊗	⊗
NCM	●	●	●		●	●	⊗	⊗	⊗	⊗	⊗	⊗
SUST	●	●	⊗	⊗	⊗	●	●	⊗	⊗	⊗	⊗	●
AW	⊗	⊗	⊗	●	●	⊗	⊗	⊗	⊗	⊗	⊗	⊗
PCM	●	●	●	●	●		●	●	●		●	●
NE	⊗	⊗	●	●	●	●	●	⊗	⊗	●	●	⊗
ACM	⊗	⊗	⊗	⊗	⊗	⊗	⊗			●	●	●
Consistency	0.981	0.9794	0.9824	0.9772	0.9808	0.9849	0.9833	0.9552	0.9564	0.9541	0.9535	0.9588
Raw coverage	0.2119	0.2121	0.2011	0.2126	0.2016	0.2318	0.18	0.228	0.2275	0.2549	0.2537	0.2265
Unique coverage	0.0052	0.0055	0.0135	0.011	0.0047	0.0317	0.0075	0.0139	0.0096	0.0077	0.008	0.0256
Overall solution consistency	0.9655							0.9317				
Overall Solution coverage	0.3642							0.3370				

Note: VICMR: visit intention for a cultured meat restaurant, SCCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat.
Note: ● indicate the presence of a causal condition; ⊗ indicate absence.

was conducted to identifying the related antecedents that are necessary to promote the desired outcome, such as visit intention for a cultured meat restaurant. The optional necessity consistency cut-off of 0.9, which was suggested by Tóth et al. (2015), was utilized to analyze the necessary conditions. The scores of the antecedent condition consistency were all unfortunately below the threshold, ranging from 0.591 to 0.799, which means that a single factor is not sufficient to be a necessary condition for the desired outcome. The findings contradict the

Table 9
The results of necessary condition analysis.

Antecedent condition	Outcome condition	Coverage
	VICMR	
	Consistency	
SCM	0.7523	0.7409
~SCM	0.5665	0.7469
QCM	0.7606	0.7632
~QCM	0.5963	0.7672
NCM	0.7610	0.7464
~NCM	0.5472	0.7254
SUST	0.7480	0.7370
~SUST	0.5610	0.7393
AW	0.7368	0.7304
~AW	0.5573	0.7284
PCM	0.7544	0.7449
~PCM	0.5445	0.7154
NE	0.7171	0.7179
~NE	0.5817	0.7506
ACM	0.7297	0.7226
~ACM	0.6060	0.7932

Note: VICMR: visit intention for a cultured meat restaurant, SCM: safety of consuming cultured meat, QCM: quality of cultured meat, NCM: nutrients of cultured meat, SUST: sustainability, AW: animal welfare, PCM: price of cultured meat, NE: new experience, ACM: aversion to cultured meat. ~ indicates negation condition. Necessary antecedent condition for achieving outcome condition (Consistency >0.9).

theoretical premise proposed by Dul (2016), which argues that certain necessary conditions can singularly determine business outcomes. However, this discrepancy does not undermine the relevance of necessary condition analysis; rather, it highlights the need to consider multiple interacting factors in consumer decision-making processes.

5. Discussion

The findings of this study offer valuable insights into consumer perceptions and behavioral intentions toward cultured meat restaurants. By employing both IPA and fsQCA, this research highlights the complex interplay of factors such as safety, quality, ethical values, and novelty in shaping visit intentions. While previous studies have primarily focused on a limited set of attributes, our results demonstrate that consumer acceptance is influenced by a broader combination of elements, including price, food neophobia, and nutritional value. The identification of key causal recipes through fsQCA further emphasizes that no single factor dominates consumer decision-making; rather, a combination of factors enhances or inhibits their willingness to visit cultured meat restaurants.

As the global cultured meat industry continues to evolve, it is essential to consider ethical concerns and potential risks related to economically motivated adulteration (EMA) in food labeling and advertising. Different countries have established guidelines to ensure transparency and prevent misleading practices in the food industry. For example, the European Union (EU) enforces strict regulations requiring that food labels provide accurate information without misleading consumers, particularly regarding nutritional claims and ethical sourcing (Paganizza, 2020). Similarly, the U.S. Food and Drug Administration (FDA) and the Food Safety Modernization Act (FSMA) emphasize the need for hazard analysis and risk-based preventive controls to mitigate adulteration risks (Golan et al., 2004).

Additionally, the role of cultural and religious dietary restrictions should not be overlooked in understanding consumer preferences. Different ethnic and religious groups have specific dietary laws that influence their acceptance of cultured meat. For instance, adherence to Halal food standards under the Organization of Islamic Cooperation (OIC) Standards and Metrology Institute for Islamic Countries (SMIIC) guidelines is a critical consideration for Muslim consumers, while Buddhist and Taoist traditions may also impose dietary restrictions (Bonne and Verbeke, 2008). Failure to address these concerns may limit market reach and consumer trust. Therefore, transparent labeling, collaboration with religious authorities, and clear regulatory guidelines will be essential in fostering widespread acceptance across diverse consumer groups.

Notably, ethical concerns and sustainability, often emphasized in cultured meat discourse, were found to be secondary to more immediate consumer considerations like taste and affordability. These findings align with existing literature on food innovation adoption while also offering practical guidance for industry stakeholders seeking to develop effective marketing and operational strategies for cultured meat establishments.

5.1. Theoretical implications

This study significantly extends the attributes associated with cultured meat via a comprehensive qualitative and quantitative approach. Our study systematically identifies all relevant factors that influence consumers when selecting a cultured meat restaurant, which is unlike the previous research that typically focused on three to four factors. The findings reveal a dual sentiment among customers, such as curiosity and aversion towards consuming cultured meat, a trend that is consistent with the existing research in food services (Fuste-Forne, 2021). Cultured meat emerges as a novel food that triggers these types of reactions. This addresses food neophobia, the rejection of unfamiliar foods. Understanding the perspectives of customers on cultured meat is crucial in predicting behavior and offering strategic insights for industry development.

This study pioneers the application of IPA analysis to classify attributes into quadrants, which is a novel approach in cultured meat research. Identifying strengths and opportunities, such as controlled fat and price, in the keep up the good work quadrant is crucial. Focusing on attributes, such as preventing infectious disease, taste and flavor, and food neophobia in the 4th quadrant is imperative. These quadrant classifications provide strategic guidance for the development of the cultured meat industry, which emphasizes the significance of the quadrant 4 factors.

The adoption of the SEM and fsQCA over a single net effect analysis enhances the depth of insights. The net effect analysis confirms the significant impact of cultured meat attributes on visit intentions, and the fsQCA results highlight nuanced outcomes. The influence of new experiences on visit intentions is not statistically significant, which is possibly due to consumers not yet perceiving cultured meat as a novel food. The SEM underlines the interconnectedness of environmental awareness, feelings about cultured meat, quality, nutrients, and price in shaping the cultured meat industry's trajectory (Liu et al., 2018; Jung et al., 2022; Siddiqui et al., 2022). The causal recipes provide more insightful knowledge to investigate the behavior intentions of consumers. The sufficient configurations indicated that the high visit intention for a cultured meat restaurant is powerfully dependent on the combined attributes of cultured meat. The price of cultured meat, aversion to cultured meat, the nutrients of cultured meat, and new experiences in food are regarded as the most influential elements in the causal recipes in the 7 causal models, and the rest of the elements, such as safety of cultured meat, quality of cultured meat, and sustainability of culture meat are also elements with a higher frequency in the causal recipes. The higher the customer's rating on these six factors, which include SCM, QCM, NCM, SUST, PCM and NE, the higher the willingness overall to visit cultured meat restaurants. On the other hand, if the level of customer evaluation for cultured meat aversion is low, there is a high intention to visit. However, the animal welfare factor was excluded, which is probably because the consumers perceive livestock as a commodity and do not judge whether it is ethical to slaughter livestock. Nevertheless, the comprehensive configurations convey a signal that seven factors, which include SCM, QCM, NCM, SUST, PCM, NE, and

ACM, could be the focus in the casual recipes for visit intention for a cultured meat restaurant objectively speaking. These results show that the merits of the cultured meat industry for various environments and animal welfare are important, but price, quality, nutritional aspects, and aversion to cultured meat affect visit intention more. It can once again be confirmed that the customer acceptance of cultured meat comes first (Verbeke et al., 2015).

The findings of the necessary condition analysis in this study do not fully align with Dul's (2016) assertion that a single necessary condition critically influences expected outcomes in business management. While this does not undermine the value of necessary condition analysis, it suggests that a single antecedent may not hold the same necessity across different industrial contexts. Instead, the combined effects of multiple antecedents may enhance the adequacy and necessity of any single factor. Since no individual factor meets the necessity threshold, businesses should adopt a multifaceted approach by integrating key attributes such as taste, food safety, and environmental benefits to enhance consumer appeal. Furthermore, the findings indicate that the adoption of cultured meat may challenge existing food norms and values, necessitating targeted consumer education and transparent communication strategies. Rather than relying on a single persuasive element, businesses should implement a holistic strategy that addresses multiple consumer concerns to effectively drive visit intention. Overall, this study provides insights into how the emergence of cultured meat may disrupt traditional food systems and practices while reshaping established consumer perceptions and behaviors.

5.2. Practical implications

The growing consumer interest in cultured meat signals an emerging market trend; however, a deeper understanding of this innovative food source is essential, as revealed by the interviews and survey results. Cultured meat remains a novel concept to the public, and systematic, strategic communication is necessary to reduce consumer resistance. **First**, targeted informational campaigns should emphasize the environmental and ethical advantages of cultured meat, aiming to raise consumer awareness. The use of neutral and positive terminology, such as "cell-based meat" or "sustainable alternative meat," rather than "artificial meat," is imperative to prevent the reinforcement of negative perceptions. Furthermore, transparent communication about the production processes should be prioritized, utilizing platforms such as documentaries, social media, and live factory tours. By presenting verifiable data demonstrating that cultured meat production requires fewer resources and supports animal welfare compared to conventional meat production (Verbeke et al., 2015; Bekker et al., 2017; Bryant et al., 2020), consumers' understanding and acceptance can be significantly enhanced. Notably, the resistance to cultured meat, highlighted in both interviews and IPA analysis, is often rooted in consumer ignorance regarding the production process (Tomiyama et al., 2020). Overcoming this knowledge gap is essential, particularly given the psychological aversion triggered by the term "artificial meat." To address this, food industry managers must adopt comprehensive strategies to alleviate resistance. Initial marketing efforts can target curiosity-driven consumer segments, gradually cultivating a positive perception through word-of-mouth promotion and firsthand exposure. Previous studies indicate that positive consumer experiences significantly influence revisit intentions and word-of-mouth behavior (Han et al., 2009; Kim et al., 2020). Additionally, introducing high-end menu offerings featuring cultured meat in restaurants catering to environmentally conscious high-income consumers could further boost curiosity and behavioral intent.

Price is another critical factor influencing consumer choices related to cultured meat. Ensuring price competitiveness is essential for broad market adoption. To foster market penetration, it is recommended that high-end restaurants initially offer cultured meat products to high-income consumers and those with environmental concerns. Over time, as the production infrastructure scales, costs can be progressively reduced. The industry should focus on achieving cost reductions through technological innovations and collaborative efforts with large food companies and governmental stakeholders. Strategic investments, including partnerships with major corporations and government-funded grants, can help lower production costs, making cultured meat more accessible to a larger segment of the population (Bryant et al., 2020; Choudhury et al., 2020).

According to IPA results, several attributes of cultured meat fall within the "concentrate here" quadrant, indicating areas requiring immediate improvement. The sensory characteristics of cultured meat, such as texture, taste, and aroma, are fundamental to its appeal. If these attributes do not meet consumer expectations, acceptance may be hindered (Ryu and Han, 2010). Therefore, ongoing investment in technological advancements to improve the quality of cultured meat is essential. Additionally, the introduction of hybrid products, blending cultured meat with traditional or plant-based meats, could reduce heterogeneity and increase overall acceptance. Establishing experiential spaces in urban centers, where consumers can directly engage with cultured meat products would also help reduce psychological resistance. These spaces could offer tasting experiences while providing insights into the production process, thereby fostering greater consumer understanding and trust. Organizing workshops, where chefs and scientists collaborate to introduce cultured meat, could further contribute to building consumer confidence and familiarity with the product.

According to the IPA results, for attributes related to consumer concerns, such as unnatural perception, food safety, health safety, and food neophobia, businesses should implement clear and transparent communication strategies to educate consumers. The expansion of the cultured meat industry requires the establishment of clear retail standards for classifying products based on their raw material origins, including red meat, white meat, mammalian meat, avian meat, wildlife and game meat, and aquatic meat. Standardized industry definitions will enhance regulatory clarity and consumer trust, ensuring transparent categorization in the market. Ethical concerns also require attention, particularly regarding transparency in species origin. Misrepresentation or failure to disclose species information may lead to ethical dilemmas, and there is a potential risk of mixing low-cost raw materials with premium cultured meat ingredients, which could erode consumer confidence. To mitigate these risks, government agencies should enforce strict labeling regulations, industry associations should develop clear guidelines, and companies must adhere to ethical sourcing practices. Additionally, consumer advocacy groups can play a role in monitoring compliance and educating the public about cultured meat transparency. Another critical aspect is the nutritional composition of cultured meat, including both micronutrients and macronutrients. Establishing precise nutritional standards within the food manufacturing sector will help ensure product consistency, enhance consumer confidence, and support the integration of cultured meat into mainstream diets. Implementing these measures collectively will help build trust and credibility in the cultured meat industry.

Finally, the food service industry stands to benefit from incorporating cultured meat into their offerings, aligning both sustainability objectives and profitability goals. High-end restaurants and hotels could develop exclusive menus featuring cultured meat, targeting consumers who prioritize environmental sustainability and ethical consumption. Cultured meat can also help reduce food waste in the food service sector due to its ability to be produced in controlled, precise quantities, thus optimizing resource use and minimizing ingredient over-ordering. Additionally, adopting cultured meat can enable the food service industry to reduce its carbon footprint and meet sustainability goals by decreasing reliance on traditional livestock farming. As cultured meat becomes more widely available, it will be crucial to educate restaurant staff to respond effectively to consumer inquiries and concerns. Promoting menus that feature sustainable cultured meat options can enhance a restaurant's eco-friendly reputation and attract

environmentally conscious consumers.

6. Limitations, future research directions, and conclusions

The current study on cultured meat is constrained by several limitations, which is due to its status as a nascent and evolving technology. First, while the research assumes the safety of cultured meat, uncertainties regarding its long-term health effects remain. Current findings suggest safety (Bryant et al., 2020), but comprehensive longitudinal studies are needed to confirm its health implications. Second, the study's geographic and cultural focus on China limits global generalizability. Cultural barriers in regions where traditional meat consumption is deeply ingrained may affect adoption. Future research should explore cultural variations in consumer acceptance and the role of social norms in shaping attitudes toward cultured meat. Additionally, many consumers remain unfamiliar with cultured meat, which may skew perceptions. Comparative research on experienced and non-experienced consumers would help address knowledge gaps and misconceptions. Investigating sensory attributes (e.g., taste, texture) in real-world settings could provide actionable insights for product development. Lastly, the evolving regulatory landscape plays a pivotal role in commercialization. Future studies should examine the impact of policy frameworks on consumer trust and market readiness. While the regulatory stabilization of cultured meat is an urgent priority at present, once regulations are effectively implemented and stabilized, there will be a need to conduct studies on the safety, quality, and trade implications of cultured meat in countries that import meat from China, including neighboring Asian nations. This study does not delve deeply into these aspects, representing a limitation that future research should address. Addressing these limitations will advance understanding and foster sustainable development within the food service and hospitality industry.

CRediT authorship contribution statement

Lanji Quan: Writing – original draft. **Heesup Han:** Supervision, Project administration.

Funding

Doctoral research start-up fund project (Project number 2024XBS04), YanBian University, Jilin Pro., China.

Declaration of competing interest

none.

Data availability

Data will be made available on request.

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